Coding & Solution

Date	15 July 2026
Team ID	SWTID-2026-8038
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details

Repository Link / URL	https://github.com/samianushna-cmyk/credit_card_approval/tree/main/Credit_card_approval_system-main
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask

Key Features Implemented	• Credit card approval prediction using Random Forest Classifier. • Data preprocessing and feature engineering. • Label encoding of categorical features. • User-friendly Flask web application for applicant data entry. • Real-time prediction of **Approved** or **Rejected** status. • Model evaluation using Accuracy, Precision, Recall, F1Score, ROC-AUC, and Confusion Matrix. • Feature importance analysis. • Responsive web interface using Bootstrap.

Pending / Incomplete Features	• User authentication (Login/Registration). • Prediction history storage in a database. • Explainable AI (SHAP/LIME) for prediction explanations. • Model retraining with newly collected data. • Cloud database integration and deployment enhancements.

Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using `pip install -r requirements.txt`. 3. Ensure `credit _card _model .pkl` is available in the project directory. 4. Run the application using `python app.py`. 5. Enter applicant details and click **Predict Credit Approval** to view the result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	YES
2	Meaningful variable and function names are used	YES
3	Code includes comments / documentation where necessary	YES
4	Error handling is implemented for critical operations	YES
5	The application runs without critical errors	YES
6	Code is committed to a version control repository	YES

Additional Notes / Comments:

The project follows standard coding practices with proper indentation, meaningful variable and function names, modular programming, and exception handling. The code is organized into separate frontend and backend components, making it easy to understand, maintain, and extend.